Assignment 3: Software-Defined Networking

COL7524: Advanced Networks
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We used the two supplied topologies without changing their links, delays or host connections. The experiments below compare reactive and proactive rules, shortest-delay routing, and routing that reacts to link load. All results come from the saved Mininet runs included with this report.

1. Reactive and proactive flow management

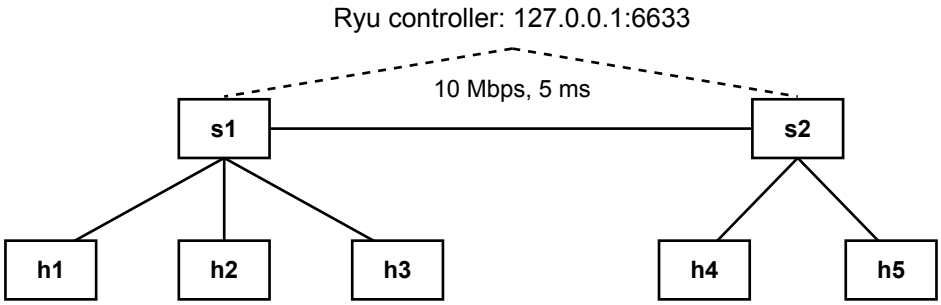


Figure 1. The two-switch topology used in Part 1. Dashed lines show controller connections.

The reactive controller learns the incoming port for each source MAC. An unknown destination is flooded; once its port is known, the controller installs a forwarding rule. The proactive controller already knows where all five hosts are connected, so it installs five destination rules and one broadcast rule on each switch before testing begins.

For each controller, we started a clean network, saved the initial tables, ran pingAll, and sent ten pings from h1 to h5. We then saved the final tables and counted the reactive controller's Packet-In log entries. The count covers the whole test sequence, including ARP and other non-LLDP Ethernet traffic handled by that application.

Measure	Reactive	Proactive
pingAll loss	0%	0%
Forwarding Packet-In events	105	0
Mean h1-h5 ping RTT (ms)	10.140	10.131
Initial rules per switch	1 table-miss rule	6 forwarding rules

The warmed-up RTTs are close because both switches then forward locally. The main difference is at setup: reactive forwarding needs controller decisions for initial traffic but installs rules only when needed. Proactive forwarding avoids that initial controller work, but needs advance knowledge and installs rules even for unused destinations. First-packet delay was not measured separately here.

The proactive count is supported by its flow tables: there is no output-to-controller rule. The before/after tables, Ryu output and successful ping summaries on pages 2-3 show this difference.

1.1 Reactive forwarding: recorded evidence

Source: p1/results/p1-reactive.txt and p1/results/p1-reactive-controller.log. Flow rows retain packet/byte counters, priority, match and action; only the cookie, duration and table number are omitted.

Commands used for controller and switch inspection

```
.venv/bin/ryu-manager --ofp-tcp-listen-port 6633 p1/reactive.py
sudo python3 p1/p1_topo.py
ovs-ofctl -O OpenFlow13 dump-flows s1
ovs-ofctl -O OpenFlow13 dump-flows s2
```

The driver takes the first table dump before pingAll and the ten h1-h5 pings, then takes the final dump after those tests. It runs the same controller/topology and invokes these flow-table commands through Mininet.

Before tests: only the table-miss rule on each switch

```
s1: n_packets=30, n_bytes=2580, priority=0 actions=CONTROLLER:65535
s2: n_packets=30, n_bytes=2580, priority=0 actions=CONTROLLER:65535
```

Selected Ryu output

```
Switch 1 connected
Switch 2 connected
Packet-In: switch=1 src=00:00:00:00:00:01 dst=00:00:00:00:00:05 out=4
Packet-In: switch=2 src=00:00:00:00:00:01 dst=00:00:00:00:00:05 out=2
```

After tests: h1/h5 forwarding rules and their packet counters

```
s1: n_packets=13, n_bytes=1218, priority=1,in_port="s1-eth4",dl_dst=00:00:00:00:00:01 actions=output:"s1-eth1"
s1: n_packets=9, n_bytes=882, priority=1,in_port="s1-eth1",dl_dst=00:00:00:00:00:05 actions=output:"s1-eth4"
s2: n_packets=10, n_bytes=980, priority=1,in_port="s2-eth2",dl_dst=00:00:00:00:00:01 actions=output:"s2-eth3"
s2: n_packets=13, n_bytes=1274, priority=1,in_port="s2-eth3",dl_dst=00:00:00:00:00:05 actions=output:"s2-eth2"
```

Connectivity and controller-workload results

```
PINGALL loss percent: 0.0
10 packets transmitted, 10 received, 0% packet loss, time 1807ms
rtt min/avg/max/mdev = 10.075/10.140/10.253/0.045 ms
Logged forwarding Packet-In events: 105
```

Initially, unmatched packets go to Ryu. The Packet-In lines show h1-to-h5 packets reaching the controller on both switches. The final tables then contain priority-1 unicast rules with nonzero counters: h1-to-h5 uses s1 port 4 and s2 port 2; replies use s2 port 3 and s1 port 1.

The initial table-miss counters are already nonzero because hosts generate background Ethernet traffic before the controlled ping tests. There are still no learned unicast rules in that first snapshot. The total Packet-In count includes that startup traffic, pingAll and the measured pings.

1.2 Proactive forwarding: recorded evidence

Source: p1/results/p1-proactive.txt and p1/results/p1-proactive-controller.log. Flow rows retain packet/byte counters, priority, match and action; only the cookie, duration and table number are omitted.

Commands used for controller and switch inspection

```
.venv/bin/ryu-manager --ofp-tcp-listen-port 6633 p1/proactive.py
sudo python3 p1/p1_topo.py
ovs-ofctl -O OpenFlow13 dump-flows s1
ovs-ofctl -O OpenFlow13 dump-flows s2
```

The driver takes the first table dump before pingAll and the ten h1-h5 pings, then takes the final dump after those tests. It runs the same controller/topology and invokes these flow-table commands through Mininet.

Before tests: destination rules and broadcast rule already present

```
s1: n_packets=0, n_bytes=0, priority=10,dl_dst=00:00:00:00:00:01 actions=output:"s1-eth1"
s1: n_packets=0, n_bytes=0, priority=10,dl_dst=00:00:00:00:00:05 actions=output:"s1-eth4"
s1: n_packets=0, n_bytes=0, priority=5,dl_dst=ff:ff:ff:ff:ff:ff actions=FLOOD
s2: n_packets=0, n_bytes=0, priority=10,dl_dst=00:00:00:00:00:01 actions=output:"s2-eth3"
s2: n_packets=0, n_bytes=0, priority=10,dl_dst=00:00:00:00:00:05 actions=output:"s2-eth2"
s2: n_packets=0, n_bytes=0, priority=5,dl_dst=ff:ff:ff:ff:ff:ff actions=FLOOD
```

Selected Ryu output

```
Switch 1 connected
Installed: switch=1 dst=h5 out=4
All proactive rules installed on switch 1
Switch 2 connected
Installed: switch=2 dst=h5 out=2
All proactive rules installed on switch 2
```

After tests: h1/h5 forwarding rules and their packet counters

```
s1: n_packets=20, n_bytes=1736, priority=10,dl_dst=00:00:00:00:00:01 actions=output:"s1-eth1"
s1: n_packets=14, n_bytes=1372, priority=10,dl_dst=00:00:00:00:00:05 actions=output:"s1-eth4"
s2: n_packets=14, n_bytes=1260, priority=10,dl_dst=00:00:00:00:00:01 actions=output:"s2-eth3"
s2: n_packets=16, n_bytes=1568, priority=10,dl_dst=00:00:00:00:00:05 actions=output:"s2-eth2"
```

Connectivity and controller-workload results

```
PINGALL loss percent: 0.0
10 packets transmitted, 10 received, 0% packet loss, time 1804ms
rtt min/avg/max/mdev = 10.108/10.131/10.164/0.020 ms
Logged forwarding Packet-In events: 0
```

The h5 rules exist before testing with zero packet counters. Afterwards, those same matches and output ports have nonzero counters: s1 sends h5 traffic over port 4, and s2 sends it to host port 2. Rules for h1 carry replies in the other direction. This shows that the preinstalled rules actually forwarded packets.

Both switches also have a broadcast rule before the test. There is no CONTROLLER action in the proactive tables, so normal forwarding does not need Packet-In handling. The two successful ping checks provide the connectivity evidence alongside the installation log.

2. Link-metric-based path selection

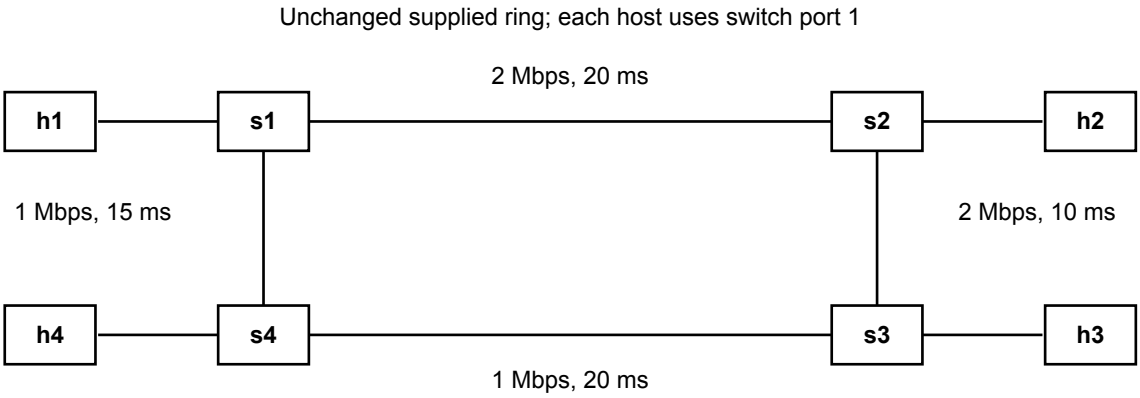


Figure 2. Parts 2 and 3 use this four-switch ring. Link labels give bandwidth and delay.

We chose delay as the static metric. The controller in p2/routing.py stores the ring in a NetworkX graph and uses the sum of link delays as path cost. The host MAC addresses and switch ports are fixed to the supplied topology: h1-h4 connect to port 1 of s1-s4. This keeps the implementation small; automatic topology or host discovery is not needed for this fixed experiment.

For each new source/destination MAC pair and EtherType, the controller finds the lowest-cost path and installs OpenFlow 1.3 rules along it. Reply traffic independently installs the reverse direction. Rules expire after five idle seconds. Broadcast ARP is sent directly to the other host ports by Ryu, rather than flooded around the ring, so there is no broadcast loop.

Test	Cost through s2	Cost through s4	Chosen h1-h3 route
Normal costs	20 + 10 = 30 ms	15 + 20 = 35 ms	s1 -> s2 -> s3
DELAY12=40	40 + 10 = 50 ms	15 + 20 = 35 ms	s1 -> s4 -> s3

In the second test, DELAY12=40 changes only the controller's s1-s2 cost. The physical topology stays unchanged. This is a link-cost-change test: the controller should prefer the other route because its total cost is now smaller.

Measured result	Normal costs	Changed cost
pingAll loss	0%	0%
Mean h1-h3 RTT (ms)	60.865	70.887

Selected controller log lines:

```
PATH 1 -> 3: [1, 2, 3] adaptive=False
PATH 3 -> 1: [3, 2, 1] adaptive=False
PATH 1 -> 3: [1, 4, 3] adaptive=False
PATH 3 -> 1: [3, 4, 1] adaptive=False
```

The unloaded round-trip delays are close to twice the physical one-way path delay: about 60 ms through s2 and 70 ms through s4. The cost change therefore selects the expected path, and the measured RTT follows the physical route. Both forward and reverse paths are present in the logs; pingAll also checks every host pair in both directions.

2.1 Route selection: recorded evidence

Source: p2/results/p2-base.txt, p2/results/p2-changed.txt and their controller logs. The selected flow rows retain counters, timeout, match and action; cookie, duration and table number are omitted.

Before traffic: table-miss rules on the four switches

s1: n_packets=6, n_bytes=516, priority=0 actions=CONTROLLER:65535

s2: n_packets=6, n_bytes=516, priority=0 actions=CONTROLLER:65535

s3: n_packets=5, n_bytes=430, priority=0 actions=CONTROLLER:65535

s4: n_packets=5, n_bytes=430, priority=0 actions=CONTROLLER:65535

Normal costs: Ryu forward and reverse paths

PATH 1 -> 3: [1, 2, 3] adaptive=False

PATH 3 -> 1: [3, 2, 1] adaptive=False

Normal costs: final h1-to-h3 IPv4 rules

s1: n_packets=8, n_bytes=784, idle_timeout=5, priority=10,ip,dl_src=00:00:00:00:00:01,dl_dst=00:00:00:00:00:03 actions=output:"s1-eth2"

s2: n_packets=8, n_bytes=784, idle_timeout=5, priority=10,ip,dl_src=00:00:00:00:00:01,dl_dst=00:00:00:00:00:03 actions=output:"s2-eth3"

s3: n_packets=8, n_bytes=784, idle_timeout=5, priority=10,ip,dl_src=00:00:00:00:00:01,dl_dst=00:00:00:00:00:03 actions=output:"s3-eth1"

Normal costs: connectivity and h1-h3 ping

PINGALL loss percent: 0.0

10 packets transmitted, 10 received, 0% packet loss, time 1805ms

rtt min/avg/max/mdev = 60.131/60.865/66.856/1.996 ms

DELAY12=40: Ryu forward and reverse paths

PATH 1 -> 3: [1, 4, 3] adaptive=False

PATH 3 -> 1: [3, 4, 1] adaptive=False

DELAY12=40: final h1-to-h3 IPv4 rules

s1: n_packets=8, n_bytes=784, idle_timeout=5, priority=10,ip,dl_src=00:00:00:00:00:01,dl_dst=00:00:00:00:00:03 actions=output:"s1-eth3"

s3: n_packets=8, n_bytes=784, idle_timeout=5, priority=10,ip,dl_src=00:00:00:00:00:01,dl_dst=00:00:00:00:00:03 actions=output:"s3-eth1"

s4: n_packets=8, n_bytes=784, idle_timeout=5, priority=10,ip,dl_src=00:00:00:00:00:01,dl_dst=00:00:00:00:00:03 actions=output:"s4-eth2"

DELAY12=40: connectivity and h1-h3 ping

PINGALL loss percent: 0.0

10 packets transmitted, 10 received, 0% packet loss, time 1804ms

rtt min/avg/max/mdev = 70.158/70.887/77.183/2.098 ms

The normal-cost rules send packets from s1 port 2 to s2 port 3 and then s3 port 1. After changing the s1-s2 cost, s1 uses port 3 instead, s4 forwards through port 2, and s3 still delivers through host port 1. Each rule has a nonzero packet counter. The reverse PATH entries and zero-loss pings confirm that replies work too.

3. Congestion-aware routing

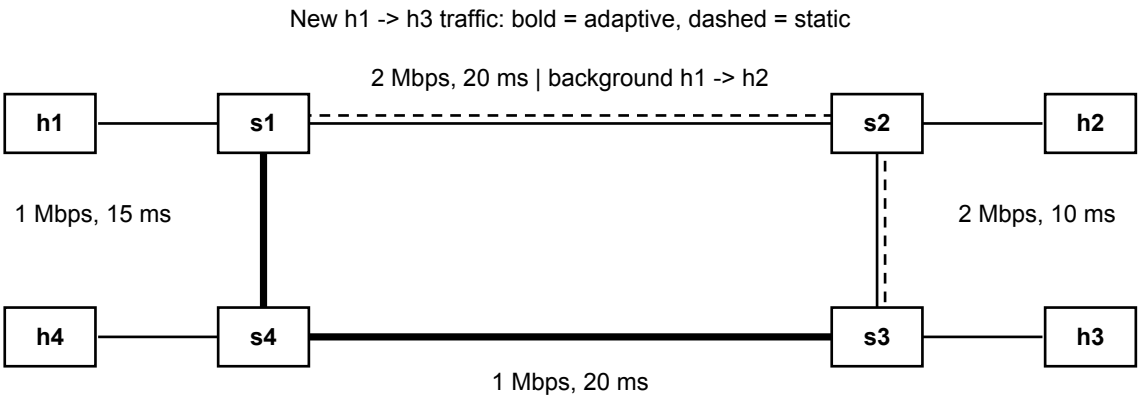


Figure 3. The background flow loads s1-s2. The later h1-h3 traffic can use the lower route.

The Part 3 controller is in p3/routing.py. It extends the static routing logic with link-utilization measurements; CONGESTION=0 selects the static comparison mode. Ryu requests port statistics once per second. For each inter-switch port we calculate utilization from the change in transmitted bytes:

Utilization rule

utilization = 8 x (new tx_bytes - old tx_bytes) / (elapsed seconds x link capacity in bits/s)

link utilization = max(utilization in the two directions)

if link utilization > 0.70: routing cost = link delay + 100 ms

otherwise: routing cost = link delay

We ran the same workload once with static routing and once with congestion-aware routing. After pingAll, we waited six seconds for the warm-up rules to expire. A h1-h2 UDP flow, configured for 24 seconds, used iperf -b 1.8M (about 1.89 Mbps) to load the 2 Mbps s1-s2 link. After five seconds of background traffic, we sent five h1-h3 pings and started an eight-second h1-h3 UDP flow at 600K bits/s. The script stops the background flow after this measurement. The nominal UDP rate is about 0.6 Mbps; iperf uses binary K units, so its offered rate is approximately 0.614 Mbps.

The h1-h3 host pair is new at this stage. It gets a path using the current measurements, while the active h1-h2 rules stay in place. Rules match a host pair and EtherType, so the UDP traffic reuses the path chosen for the preceding h1-h3 ping. A later flow for the same pair can choose again after five idle seconds.

Routing mode	h1-h3 route	Mean RTT (ms)	Receiver Mbps
Static	s1 -> s2 -> s3	61.711	0.480
Congestion-aware	s1 -> s4 -> s3	71.827	0.617

Receiver throughput is the mean of the first seven complete one-second server intervals in each run. We use the same window in both modes and exclude the final UDP summary/ACK, which can be delayed or lost under load. Both runs passed pingAll and the later ping test without loss.

At the adaptive h1-h3 decision, s1 port 2 utilization was 0.97, above the 0.70 threshold. The upper route therefore costs at least 130 ms, compared with 35 ms through s4. The alternate route delivered more of the foreground traffic in this run, although its propagation delay is longer.

One limitation is the one-second sampling interval: a short burst may finish before it is detected. OVS byte counters can include traffic waiting in the shaped egress queue, so the estimate may reflect offered load rather than exact wire utilization. There is no hysteresis, so paths can alternate near the threshold. We do not reroute active flows, and the fixed host-pair rules do not distinguish individual TCP or UDP sessions.

3.1 Congestion test: recorded evidence

Source: p3/results/p3-static.txt, p3/results/p3-adaptive.txt and their controller/receiver logs. The workload and routing rules are shown below.

Traffic commands (started on the indicated Mininet hosts)

```
h2: iperf -s -u -p 5001
h1: iperf -c 10.0.0.2 -u -b 1.8M -t 24 -p 5001
h3: iperf -s -u -i 1 -p 5002
h1: iperf -c 10.0.0.3 -u -b 600K -t 8 -p 5002
```

Under load: Ryu path decisions with measured utilization

```
PATH 1 -> 3: [1, 2, 3] adaptive=False utilization={'(1, 2)': 0.97, '(1, 3)': 0, '(4, 2)': 0, '(4, 3)': 0, '(3, 2)': 0, '(3, 3)': 0, '(2, 2)': 0, '(2, 3)': 0}
PATH 1 -> 3: [1, 4, 3] adaptive=True utilization={'(1, 2)': 0.97, '(1, 3)': 0, '(4, 2)': 0, '(4, 3)': 0, '(3, 2)': 0, '(3, 3)': 0, '(2, 2)': 0, '(2, 3)': 0}
```

Final s1 rules: static foreground, adaptive foreground, adaptive background

```
s1: n_packets=620, n_bytes=931784, idle_timeout=5, priority=10,ip,dl_src=00:00:00:00:00:01,dl_dst=00:00:00:00:00:03 actions=output:"s1-eth2"
s1: n_packets=428, n_bytes=641480, idle_timeout=5, priority=10,ip,dl_src=00:00:00:00:00:01,dl_dst=00:00:00:00:00:03 actions=output:"s1-eth3"
s1: n_packets=2272, n_bytes=3435264, idle_timeout=5, priority=10,ip,dl_src=00:00:00:00:00:01,dl_dst=00:00:00:00:00:02 actions=output:"s1-eth2"
```

For h1-h3, s1 changes from output port 2 to port 3. The active h1-h2 background rule still uses port 2. Thus the later host pair uses the updated measurements without moving the existing background traffic.

p3-static: first two receiver intervals

```
[ 1] 0.0000-1.0000 sec 60.3 KBytes 494 Kbits/sec 5.028 ms 0/42 (0%)
[ 1] 1.0000-2.0000 sec 58.9 KBytes 482 Kbits/sec 5.407 ms 0/41 (0%)
```

p3-adaptive: first two receiver intervals

```
[ 1] 0.0000-1.0000 sec 77.5 KBytes 635 Kbits/sec 0.015 ms 0/54 (0%)
[ 1] 1.0000-2.0000 sec 74.6 KBytes 612 Kbits/sec 0.012 ms 0/52 (0%)
```

Static h1-h3 ping

```
PINGALL loss percent: 0.0
5 packets transmitted, 5 received, 0% packet loss, time 801ms
rtt min/avg/max/mdev = 60.115/61.711/68.005/3.147 ms
```

Adaptive h1-h3 ping

```
PINGALL loss percent: 0.0
5 packets transmitted, 5 received, 0% packet loss, time 801ms
rtt min/avg/max/mdev = 70.090/71.827/78.604/3.388 ms
```

The receiver intervals show the reduced delivery rate on the shared upper link and the higher rate on the alternate route. The comparison uses seven complete intervals from each log. Queued traffic can keep arriving after the static sender stops, so its final summary/ACK is excluded.

4. Setup and run commands

Tested on Linux with Python 3.12.3, Open vSwitch 3.3.9, Ryu 4.34, Mininet and iperf 2.1.9. The supplied topologies are unchanged.

Extract the ZIP. Its top level contains p1/, p2/, p3/ and Report.pdf. Each part contains its controller, supplied topology, run.py and results/. Run the following commands from that top level:

Setup and complete experiment run

```
sudo apt install mininet openvswitch-switch iperf python3-venv
python3 -m venv .venv
.venv/bin/pip install setuptools==67.6.1 wheel
.venv/bin/pip install -r p1/requirements.txt
sudo systemctl start openvswitch-switch
sudo python3 p1/run.py "$PWD/.venv/bin/ryu-manager"
sudo python3 p2/run.py "$PWD/.venv/bin/ryu-manager"
sudo python3 p3/run.py "$PWD/.venv/bin/ryu-manager"
```

Each run.py executes the two cases for its part and replaces its results/. A failed connectivity check stops the run. Mininet runs with system Python and sudo; Ryu runs from the virtual environment. Stop other controllers before running. Use sudo mn -c after an interrupted Mininet run.

For an interactive run, start Ryu and the topology in separate terminals. All controllers listen on port 6633:

Part	Controller command after .venv/bin/ryu-manager	Topology command
1 reactive	--ofp-tcp-listen-port 6633 p1/reactive.py	sudo python3 p1/p1_topo.py
1 proactive	--ofp-tcp-listen-port 6633 p1/proactive.py	sudo python3 p1/p1_topo.py
2	--ofp-tcp-listen-port 6633 p2/routing.py	sudo python3 p2/p3_topo.py
3	--ofp-tcp-listen-port 6633 p3/routing.py	sudo python3 p3/p3_topo.py

For the changed-cost Part 2 case, prefix its controller command with DELAY12=40. For static Part 3, prefix its controller command with CONGESTION=0. Inside Mininet, use pingall and h1 ping -c 10 h3. Inspect a switch with sh ovs-ofctl -O OpenFlow13 dump-flows s1.